

Introduction Cloud Computing

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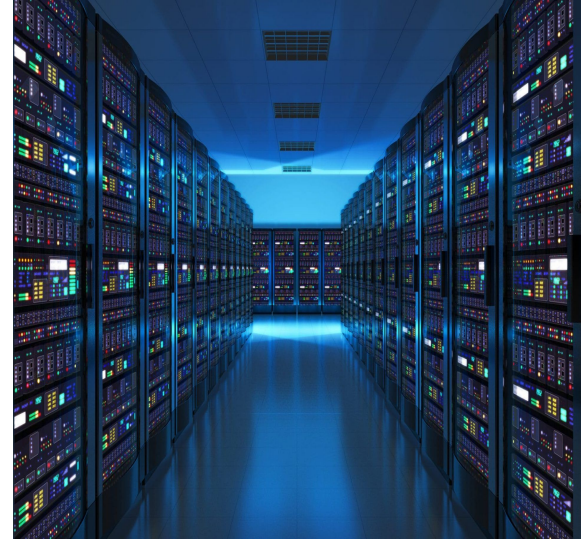
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Goals

1. General introduction
2. Types of Cloud
3. Types of Cloud services
4. Some Cloud providers

Cloud computing the idea

- **Amazon, 2002**
 - Rent to external users a part part of his facilities during period of low use
 - Creation of Amazon Web Services (AWS) initially for data storage then for compute



Cloud computing the idea

The idea basic idea, **mutualization**

- **My Car**

- Rent to people not having a car
- Reduce the Cost of ownership



- **Harvester**

- Cost a Lot of Money
- Reduce the cost when shared



Cloud Computing Definition

Set of Resources/applications/services which execute in a distributed environment (hosting center), accessible through Web standard protocols, which globally provide a service with the following characteristics:

- Pays as you Go
- Illusion of infinity of ressources
- Abstraction of the hardware infrastructure
- Mutualization between many users

Cloud Computing Definition

NIST Definition of Cloud Computing

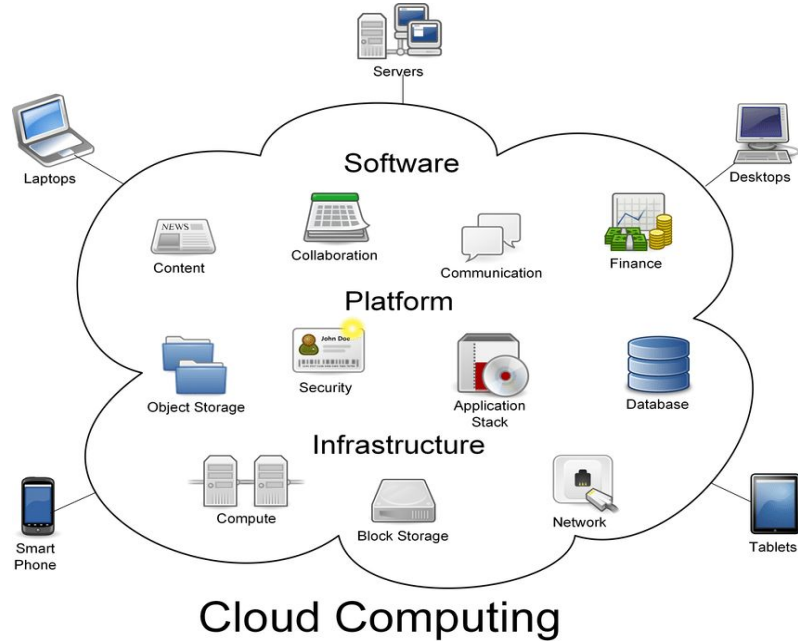
"Cloud computing is a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction."

Essential Characteristics of Cloud Computing

- **On-demand self-service:** users provision resources without human interaction with the provider.
- **Broad network access:** services are accessible through standard network mechanisms.
- **Resource pooling:** physical resources are shared between multiple users.
- **Rapid elasticity:** resources can rapidly scale up and down according to demand.
- **Measured service:** resource consumption is monitored and typically billed according to usage.

Cloud Computing Definition

General scheme is externalization of resources



Cloud computing

- Enormous **Computer data-centers (DC)** containing Commodity hardwares
- Achieve economies of scale.
 - Reduce costs of electricity, bandwidth, hardware, software and use low-cost locations
 - Lower-cost than provisioning own hardware

Cloud computing

- Cloud computing is **Utility Computing**
 - Cloud services are controlled and monitored by the cloud provider through a **pay-per-use business model**.
- An ideal cloud computing platform is:
 - Efficient in its use of resources
 - Scalable
 - Elastic
 - Self-managing
 - Highly available and accessible
 - Inter-operable and portable

Utility computing is a service provisioning model in which a service provider makes computing resources and infrastructure management available to the customer as needed, and charges them for specific usage rather than a flat rate.

Scalability vs Elasticity

- **Scalability**: ability to handle an increasing workload by adding resources.
 - **Vertical scaling (scale-up)**: use a more powerful machine.
 - **Horizontal scaling (scale-out)**: use more machines.
- **Elasticity**: ability to dynamically add and remove resources according to workload.

10 VMs → traffic increases → 50 VMs → traffic decreases → 10 VMs

Scalability = ability to grow

Elasticity = ability to adapt dynamically

Roles in the Cloud

- **Cloud provider**
 - Provides hardware infrastructures and a set of services on top
- **Cloud Clients**
 - Use resources of the Cloud
- **Cloud resellers**
 - Build and Sells a Service that rely on the Cloud.
- **Cloud developers**
 - Produce tools for the Cloud

Types of Cloud

- There are **four primary cloud deployment models** :
 - Public Cloud
 - Private Cloud
 - Community Cloud
 - Hybrid Cloud

Public Cloud

- Public clouds are owned by cloud service providers who charge for the use of cloud resources.
- Basic characteristics:
 - Shared resources and multi-tenancy
 - Leased or rented infrastructure
 - Economies of scale



Private Cloud

- The cloud infrastructure belongs to and is operated by only one organization.
- Basic characteristics :
 - Heterogeneous infrastructure
 - Customized policies
 - Dedicated resources
 - In-house infrastructure
 - End-to-end control

 EUCALYPTUS



Community Cloud

- The cloud is built to be shared by several organizations for their usage



Hybrid Cloud

- The cloud infrastructure is a composition of two or more clouds (private, community, or public) that remain unique entities but are bound together by standardized or proprietary technology that enables data and application portability.

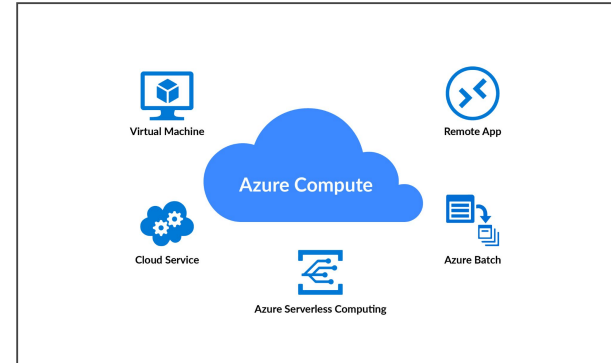
Service Classifications

- **Infrastructure as a Service (IaaS)**
- **Platform as a Service (PaaS)**
- **Software as a Service (SaaS)**



Infrastructure as a Service (IaaS)

- A type of cloud computing service that offers essential compute, storage, and networking resources on demand, on a pay-as-you-go basis
 - **Virtualization**
 - **Bare metal**



Platform as a Service (PaaS)

- A computing platform that abstracts the infrastructure, OS, and middleware to drive developer productivity



Software as a Service (SaaS)

- Run applications on a provider's on a cloud infrastructure. Applications are accessible from various client devices through a thin client interface such as a web browser.



Serverless, Function as a Service (More later)

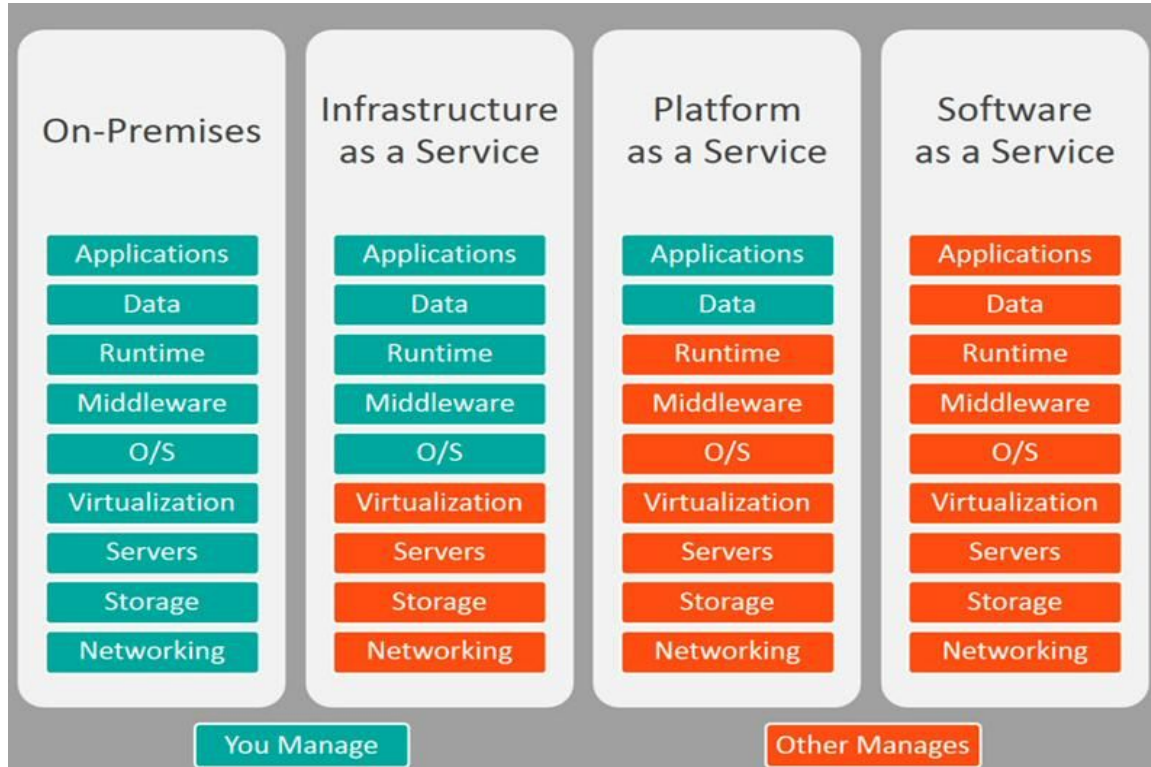
- Serverless is an event-driven or code execution where underlying infrastructure is abstracted. Applications consists of stateless functions that are spawned based on the triggering of events.



AWS Lambda

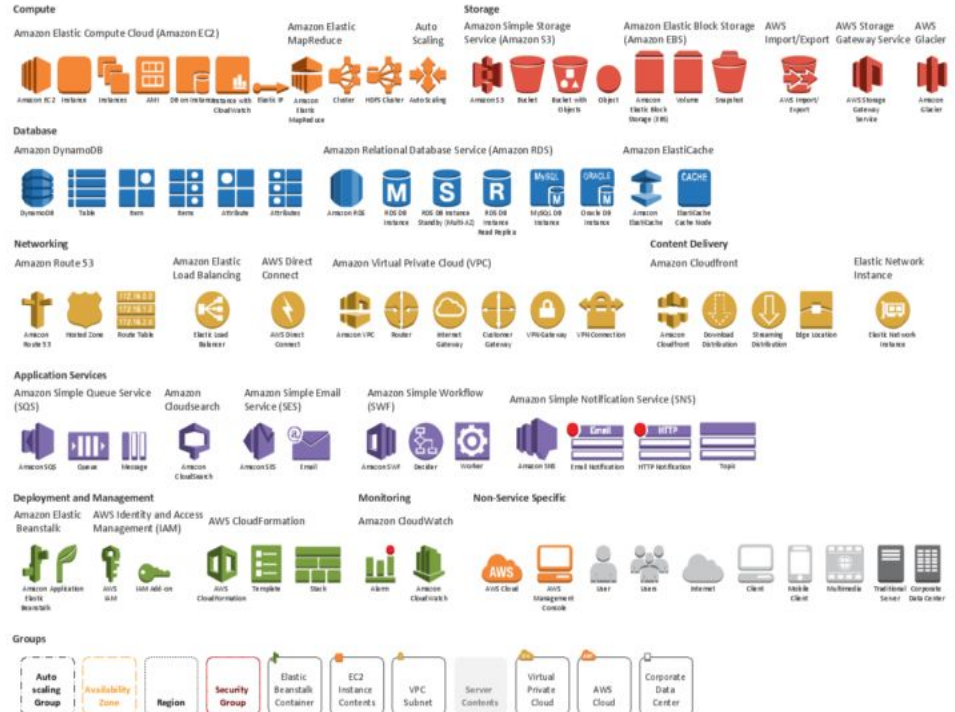


Cloud Services



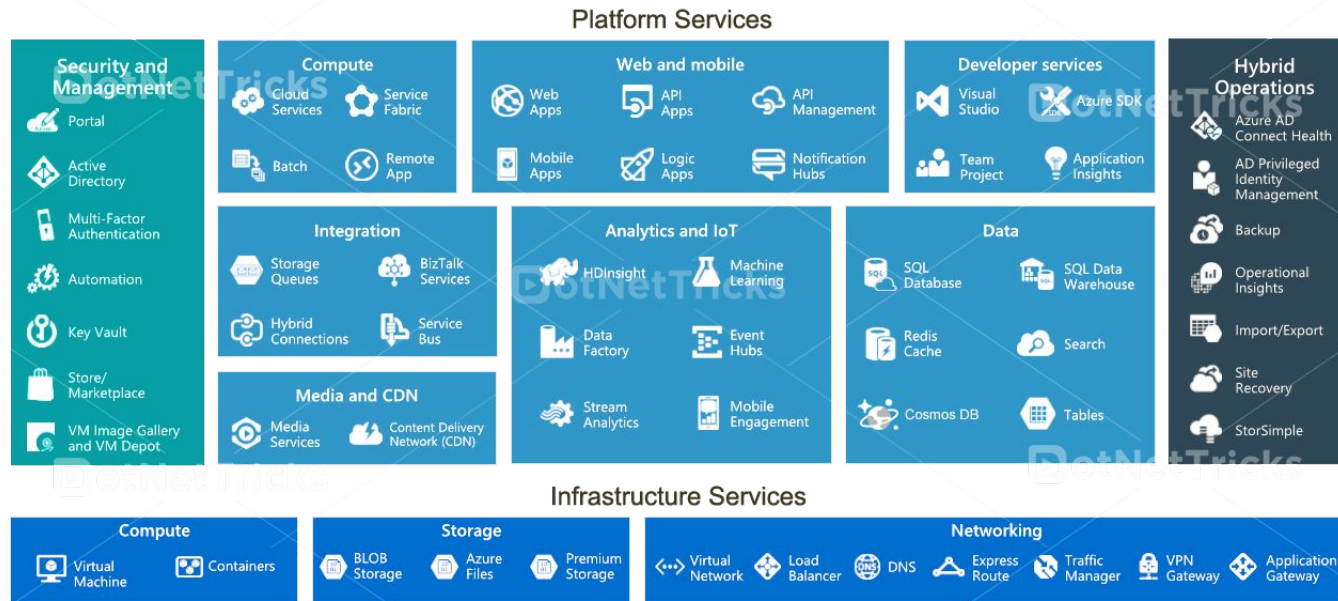
Some Cloud providers

- **Amazon AWS**
 - Public Cloud
 - Most popular cloud provider



Some Cloud providers

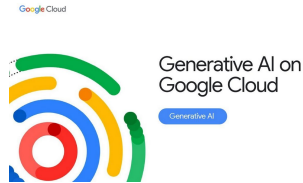
- **Microsoft Azure**
 - Public Cloud
 - Provides several services



Cloud Use Cases

Why Companies Adopt Cloud Computing **Content:**

- Access to on-demand computing resources
- Reduced infrastructure costs
- Flexibility and scalability
- Faster innovation and global deployment:



- Example: A small business can launch a global website without owning a datacenter

Data Storage and Backup

Storage and Backup:

- Cloud services: Dropbox, Google Drive, OneDrive, AWS S3
- Automatic backup and fast recovery in case of failure
- Real-time collaboration on files:
- Example: a team shares documents in Google Drive without a local server



Artificial Intelligence and Machine Learning

AI and Machine Learning in the Cloud:

- ML and AI models accessible via Cloud: AWS SageMaker, Azure ML, Google AI Platform
- Benefits: massive computing power, centralized data
- Examples: chatbots, image recognition, sales forecasting
- Example: voice recognition apps (Siri, Google Assistant)

Examples of Companies Migrating to the Cloud

Companies and Cloud Migration:

- Netflix: global streaming based on AWS
- Airbnb: scalable infrastructure on AWS
- Spotify: data storage and processing on Google Cloud
- Schneider Electric: modernization of internal systems via Azure

Data-centers (DC)

- Scaled-up/out version of machine rooms for enterprise computing
- A **large collection** of commodity components
 - PC-based servers (CPUs, DRAM, disks),
 - Ethernet networking
 - Commodity OS and software stack
 - 10s to 100s of thousands of nodes
- High-bandwidth networking (10Gbps 40Gbps 100Gbps)
- Power delivery, cooling, UPS (50 – 200MW)
- System software for DC management (centralized or distributed)
- Software that implements internet services



Data-centers (DC)

- ~500 Datacenters (w/ >10k servers)
- ~ \$30 billion Workloads for AWS 2018
- Microsoft and Google > 2,000,000 servers
- ~3% of US Power consumption



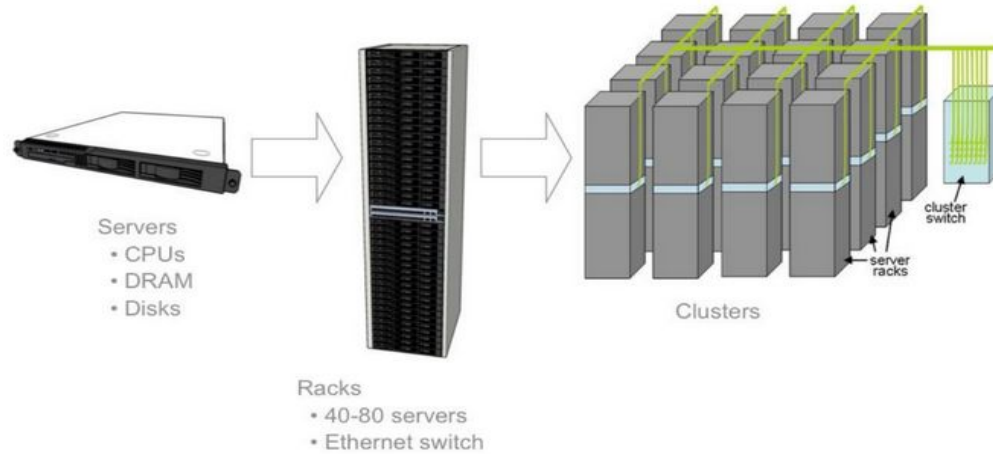
Ouvert en 2009, le data center [Azure](#) de Microsoft à Dublin représente pour l'heure une capacité de 39 000 mètres carrés de salles informatiques. L'équivalent de presque huit terrains de football. 80 salariés de [Microsoft](#) travaillent sur le site, aux côtés de quelques prestataires triés sur le volet.

Data-centers (DC)



Pour la première fois, [Google](#) lève le voile sur l'un de ses secrets les mieux gardés : ses centres de données. Ici, le datacenter de Council Bluffs, dans l'Iowa, qui s'étend sur plus de 10 000 mètres carrés. Il est notamment utilisé pour héberger le moteur recherche de Google ainsi que YouTube.

Data-centers (DC)



DataCenters

- **Compute and Memory**
 - A PC-based server (most of them x86)
 - Custom Linux OS (optimized for latency)
 - 1U, 2U, or blade form factors are popular
- **Networking**
 - 10Gbps ethernet links to rack switch
 - A few 10Gbps links to cluster level switches
- **Storage**
 - Distributed file system using disks on servers (GFS)
 - Network attached storage (NAS) devices

Data-centers (DC)

AWS Regions



Data-centers (DC)

Availability Zone (AZ): one or more physically separate datacenters within a region.

Region: geographical area containing multiple availability zones.

Deploying across multiple AZs improves availability and fault tolerance.

Cloud Computing: Benefits and Trade-offs

Benefits

- On-demand resources
- Elasticity
- Rapid provisioning
- Global infrastructure
- Managed services
- Reduced upfront investment

Trade-offs

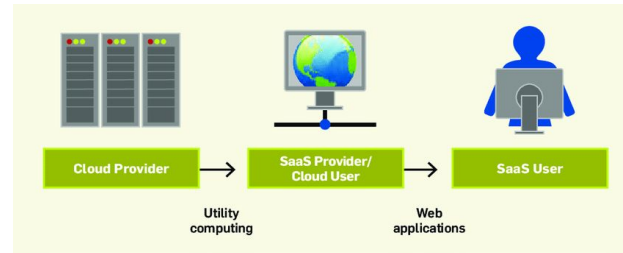
Provider dependency / vendor lock-in
Unpredictable costs
Network dependency
Data sovereignty
Security shared with provider
Less control over infrastructure

Cloud is not necessarily cheaper. Its major advantages are flexibility, elasticity and automation.

Cloud Security

Key Concepts in Cloud Security Content:

- Data encryption in transit and at rest
- Identity and access management (IAM)
- Multi-factor authentication (MFA)
- Regular security audits and compliance standards (ISO 27001, SOC 2, GDPR)
- Shared responsibility model between provider and user



- Example: Using IAM policies to restrict access to sensitive data

Cloud Security

Understanding the CLOUD Act

- The Clarifying Lawful Overseas Use of Data Act (CLOUD Act) allows US authorities to access data stored by US-based cloud providers, even if stored overseas
- Implications for privacy and cross-border data
- Importance of knowing provider policies and local regulations
- Impact on businesses: compliance and legal considerations

Some Cloud providers

- French Cloud providers
 - OVHCloud
 - Eolas
 - OutScale
 - ScaleWay

